

SGS功能安全考试题

功能安全考试题汇总

1. General FS Management（通用功能安全管理）

Q1. Which document shall manage the Functional Safety responsibilities in a development project?

以下哪个文件管理开发项目中的功能安全职责？

- A. Safety Case（安全档案）
- B. Safety Plan（安全计划）
- C. Assessment Plan（评估计划）
- D. Development Interface Agreement, DIA（开发接口协议）

Q2. Which aspects are mandatory for the specification of safety requirements?

安全需求规范中，哪些方面是必须的？

- A. Limited number（数量限制）
- B. Hierarchical structure of requirements（需求的层次结构）
- C. Formal notation（形式化表示法）
- D. Traceability of requirements（需求的可追溯性）

Q3. Which steps are typically parts of change management?

哪些步骤通常是变更管理的一部分？

- A. Impact analysis（影响分析）
- B. Assessment of change（变更的评估）
- C. Approval and documentation of change（变更的批准和记录）
- D. Change request（变更请求）

Q4. Which verification methods are highly recommended for ASIL D?

ASIL D 强烈建议采用哪些验证方法？

- A. Verification by walkthrough（通过走查进行验证）
- B. Semiformal verification（半形式化验证）
- C. Formal verification（形式化验证）
- D. Verification by inspection（通过检查进行验证）

Q5. The classification TI2 and TD2 leads to which confidence level?

分类 TI2 和 TD2 对应的置信水平是？

- A. TCL3
- B. TCL1
- C. TCL2
- D. No valid combination (没有有效的组合)

Q6. Which qualification is highly recommended for a software tool with TCL3 used in an ASIL C application?

对于 ASIL C 应用中使用的 TCL3 软件工具，强烈建议采用哪种鉴定方法？

- A. Evaluation of the tool development process (工具开发流程评估)
- B. Tool validation (软件工具确认)
- C. Qualification by confidence in use (使用中积累置信度)
- D. Formal verification (形式化验证)

Q7. A resistor is typically classified with which complexity?

电阻通常按以下哪种复杂程度分类？

- A. Complex hardware component (复杂硬件组件)
- B. Basic hardware part (简单硬件元器件)
- C. Intermediate hardware component (中等硬件组件)
- D. Intermediate hardware part (中等硬件元器件)

Q8. The proven-in-use argument is based on which information?

“在用证明”的依据基于以下什么信息？

- A. No feedback from the customer (无客户反馈)
- B. Publicly available databases (公共可用数据库)
- C. Field data (现场数据)
- D. Good feeling (良好的感觉)

2. Supporting Processes (支持过程)

Q9. What does “field observation” mean in the context of functional safety?

在功能安全语境下，“现场观察”有哪些涵义？

- A. Installation of a field-observation process focused on functional safety (建立针对功能安全的现场观察程序)
- B. Escalation management (升级管理)

C. Recording of field data / statistics (记录现场数据／统计数据)

D. Recall management (召回管理)

Q10. Which independence is recommended for the functional safety assessment of an ASIL C item?

对 ASIL C 的相关项进行功能安全评估，建议采用哪种独立性？

A. I1

B. I3

C. I0

D. I2

Q11. What are typical responsibilities of a project safety manager?

项目安全经理的典型职责是什么？

A. Calculation of development costs (开发成本核算)

B. Listing of documentation to be prepared (编制文件清单)

C. Performance of evaluations: deadlines, schedules, preparation and organization (开展评价：截止日期、日程、准备和组织)

D. Planning and tracking of the required phases in the safety lifecycle (安全生命周期所需各阶段的计划和跟踪)

E. Release for production (生产发布)

F. Initiation of an escalation process if evaluations are not successfully passed (如果评价未成功通过，启动升级流程)

G. Involvement in selection of project members (参与选择项目成员)

H. Management of internal and external interfaces, e.g. departments, customers, suppliers and TÜV (管理内部及外部接口，例如部门、客户、供应商和 TÜV)

I. Variation / tailoring of the functional-safety development process (功能安全开发流程的调整／裁剪)

Q12. What are the activities of confirmation measures?

确认措施的活动有哪些？

A. Audit of functional-safety processes during series / production development (对产品开发的功能安全流程进行审核)

B. Assessment of functional safety prior to production approval (生产批准前的功能安全评估)

C. Review of the results / work products of safety activities (对安全活动的成果／工作产品进行评审)

D. Setup of safety planning (制订安全计划)

Q13. What is the meaning of “project-independent tailoring” ?

“独立于项目的剪裁”是什么意思？

- A. Reduction of test cases (减少测试用例)
- B. Adaptation of safety-lifecycle phases in accordance with responsibilities (根据职责调整安全生命周期阶段)
- C. Use of predefined methods (使用预定义的方法)
- D. Selection of activities for cost reduction (选择降低成本的活动)

Q14. Which points are indicative of a good safety culture?

以下哪几点体现了良好的安全文化？

- A. Heavy dependence on testing at the end of the product-development cycle (严重依赖产品开发周期后期的测试)
- B. Safety is the highest priority (安全具有最高优先级)
- C. Safety and quality issues are discovered and resolved from the earliest stage of the lifecycle (安全和质量问题在产品生命周期最初阶段被发现并解决)
- D. The process assures that accountability for functional-safety-related decisions is traceable (流程确保与功能安全相关的决策责任可追溯)

Q15. Which main aspects are relevant for risk reduction in electronic systems?

主要从哪些方面降低电子系统中的风险？

- A. Control of random and systematic faults (控制随机性和系统性故障)
- B. Defense against unauthorized access (防止未授权访问)
- C. Avoidance of systematic faults (避免系统性故障)
- D. Reduction of costs (降低成本)

Q16. Non-compliance with ISO 26262 leads to what?

不符合 ISO 26262 标准将引起什么后果？

- A. Punishment by the government (政府的处罚)
- B. Unpredictable risk of product liability (不可预知的产品责任风险)
- C. Loss of confidence by the end user (丧失终端用户的信任)
- D. Stop of production (停产)

Q17. Which sentence is correct?

以下哪些描述是正确的？

- A. Application of ISO 26262 is a legal requirement concerning the approval of vehicle electronics (应用 ISO 26262 是车辆电子设备批准的法律要求)
- B. Application of ISO 26262, Edition 2, for vehicles above 3.5 t is not permissible (ISO 26262 第二版不适用于重量超过 3.5 吨的车辆)

- C. Hazards emanating from non-E/E functions are not included in the scope of ISO 26262 (由非 E/E 功能引起的危害不在 ISO 26262 考虑范围内)
- D. The defined scope of ISO 26262, Edition 2, is production passenger vehicles (ISO 26262 第二版的适用范围是量产乘用车)

Q18. What does the abbreviation “ASIL” mean?

缩写“ASIL”的含义是什么？

- A. Automotive Safety Integrity Level (汽车安全完整性等级)
- B. Automotive Security Information Level (汽车信息安全等级)
- C. A Safety Integrated Law requirement (总体安全法规要求)

Q19. What is the definition of “risk” in the functional-safety environment?

在功能安全标准中，“风险”是如何定义的？

- A. Combination of acceptable damage and the probability of such damage (可接受伤害和此种伤害可能性的组合)
- B. Combination of the probability and the extent of personal damage, injury or death (人身伤害发生的可能性与伤害程度的组合)
- C. Combination of potential material damage and personal damage, injury or death, in case of an accident (事故中可能引起的财产损失和人身伤害／死亡的组合)

3. Concept Phase (概念阶段)

Q20. What is the meaning of “semiformal notation” ?

“半形式化标记”是什么意思？

- A. Syntax and semantics are not, or are only partially, defined (语法和语义未定义或仅部分定义)
- B. Syntax and semantics are defined (语法和语义已定义)
- C. Syntax is defined, but the corresponding semantics are not, or are only partially, defined (语法已定义，而相应的语义未定义或仅部分定义)

Q21. What is the meaning of FTTI?

FTTI 是什么？

- A. Fault Tolerance Time Interval (故障容忍时间间隔)
- B. Failure Tolerance Time Interval (失效容忍时间间隔)
- C. Failure Trigger Test Interval (失效触发测试间隔)
- D. Fault Trigger Test Interval (故障触发测试间隔)
- E. Fault Tolerance Test Interval (故障容错测试间隔)

Q22. Which elements should be part of a Functional Safety Concept?

哪些要素应该是功能安全概念的一部分？

- A. Functional block diagram with functional redundancies (具有功能冗余的功能框图)
- B. Requirements to be met by risk-reducing measures of other technologies and measures (其他技术和外部措施的风险降低措施应满足的要求)
- C. Detailed description of measures to control hardware faults (控制硬件故障措施的详细说明)
- D. Requirements to be met by architectural metrics (应达到的架构指标要求)
- E. Allocation of ASIL requirements to functional elements (将 ASIL 要求分配给功能元素)
- F. Driver actions that contribute to achieving safety goals with functional redundancies (有助于通过功能冗余实现安全目标的驾驶员行动)
- G. Alarm and degradation concept (报警和降级概念)

Q23. What are the typical steps of a HARA?

HARA 的典型步骤是什么？

- A. Selection of scenarios under consideration (选择需要考虑的场景)
- B. Alignment of the risk-assessment matrix and determination of risk parameters (建立风险评估矩阵并确定风险参数)
- C. Identification of possible malfunctions of the item (识别相关项可能出现的功能故障)
- D. Verification of the analysis results (验证分析结果)
- E. Definition of responsibilities (责任的定义)
- F. Evaluation of the results (评价结果)

Q24. What is the meaning of HARA?

HARA 的含义是什么？

- A. Health Analysis and Responsibility Assessment (健康分析和责任评估)
- B. Health Analysis and Risk Assessment (健康分析和风险评估)
- C. Hazard Application and Risk Arrangement (危害应用和风险整理)
- D. Hazard Analysis and Risk Assessment (危害分析和风险评估)

Q25. What is the typical content of an item definition?

相关项定义的典型内容是什么？

- A. Safety mechanisms to be realized with E/E measures (通过电子电气措施实现的安全机制)
- B. Legal and normative requirements (法律和规范要求)
- C. Description of the function to be implemented at vehicle level (在整车层面实现的功能描述)
- D. Definition of environmental and use conditions (环境和使用条件的定义)
- E. Responsibilities in safety management (安全管理的职责)
- F. Functional block diagram presenting functional elements and their interaction, including interaction with other items (显示功能元素及其交互、包括与其他相关项交互的功能框图)
- G. External risk-reduction measures at vehicle level (整车级别的外部风险降低措施)

4. System Development (系统开发)

Q26. Which elements should be included in the Technical Safety Concept?

技术安全概念宜包括哪些要素?

- A. Description of the used safety mechanisms (所用安全机制的说明)
- B. Requirements for implementation of the warning and degradation concept (实施告警和降级概念的要求)
- C. Requirements for qualification of the system developer (对系统开发商资质的要求)
- D. Results of system analysis (系统分析的结果)
- E. Target value for PMHF (PMHF 的目标值)
- F. Technical safety requirements at system level (系统级技术安全要求)
- G. System design including safety mechanisms (包括安全机制的系统设计)
- H. Definition of activities during system development (系统开发期间活动的定义)
- I. Target values for SPFM and LFM (SPFM 和 LFM 的目标值)

Q27. A system design with a highest integrity requirement of ASIL D shall be verified by safety analysis. Which kind of safety analysis is highly recommended?

具有最高 ASIL D 完整性要求的系统设计应通过安全分析进行验证。强烈建议采用哪种安全分析?

- A. Only deductive analysis (仅演绎分析)
- B. Safety analysis is not highly recommended (不强烈建议进行安全分析)
- C. Inductive and deductive analysis (归纳和演绎分析)
- D. Only inductive analysis (仅归纳分析)

Q28. ASIL decomposition is allowed if which conditions are met?

以下哪些情形允许 ASIL 分解?

- A. The decomposed elements are independent with respect to systematic faults (分解元素对于系统性故障是独立的)
- B. The decomposed elements both fulfill ASIL B (分解元素同时满足 ASIL B)
- C. The decomposed elements are independent with respect to random hardware faults (分解元素对于随机硬件故障是独立的)
- D. The decomposed elements are homogeneously redundant (分解元素是同质冗余的)

Q29. What aspects are to be considered during system integration and test?

系统集成和测试中, 需要考虑哪些方面?

- A. Level of robustness, e.g. stress tests (鲁棒性水平, 例如压力测试)
- B. Correctness of safety-mechanism functionality (安全机制功能的正确性)

- C. Effectiveness of diagnostic coverage of hardware fault-detection mechanisms (硬件故障检测机制诊断覆盖度的有效性)
- D. Correctness of system safety-requirements implementation (系统安全要求实施的正确性)
- E. Calculation of hardware metrics (硬件指标的计算)
- F. Achievement of safety goals (安全目标的实现)

Q30. At which level is “Safety Validation” addressed in ISO 26262?

在 ISO 26262 中，“安全验证”在哪一层面进行？

- A. Vehicle level (整车层面)
- B. Hardware level (硬件层面)
- C. Software level (软件层面)
- D. System level (系统层面)

5. Software Development (软件开发)

Q31. A safe software architecture shall ensure which aspects?

安全的软件架构应确保以下哪些方面？

- A. A software structure that is verifiable (可验证的软件架构)
- B. Detailed design of software units (软件单元的详细设计)
- C. Allocation of safety-relevant functionality to subcomponents (将安全相关功能分配给子组件)
- D. Progressive refinement, e.g. software design and software-module design (逐步细化，例如软件设计和软件模块设计)
- E. Maintainable / serviceable software structure (可维护／可服务性的软件结构)

Q32. Which aspects are considered during an SW-FMEA?

在 SW-FMEA 中需要考虑哪些方面？

- A. Inconsistencies between requirements and design (需求与设计之间的不一致)
- B. Hidden and unanticipated failure modes (隐藏的和意料之外的失效模式)
- C. Dependencies within software and between software and hardware (软件内部及软硬件之间的依赖关系)
- D. System interactions (系统交互)
- E. Unstated assumptions (未说明的假设)

Q33. What is the objective of a data-flow and control-flow analysis?

数据流和控制流分析的目标是什么？

- A. Evaluation of the required test coverage (评估所需的测试覆盖率)
- B. Identification of faults in the hardware-software interface (识别硬件—软件接口故障)
- C. Identification of bad or incorrect program structures (识别不良或不正确的程序结构)

D. Verification of software safety requirements (验证软件安全要求)

Q34. According to ISO 26262-6, what percentage of MC/DC structural coverage is required for ASIL D?

根据 ISO 26262-6, ASIL D 需要达到多少 MC/DC 结构覆盖率?

- A. 80%
- B. 100%
- C. There is no target value in ISO 26262-6 (ISO 26262-6 中没有给出目标值)

Q35. Which implementation rules are highly recommended for all ASILs?

哪些实施规则是所有 ASIL 强烈建议采用的?

- A. No recursion (无递归)
- B. Initialization of variables (初始化变量)
- C. One entry point and one exit point in subprograms and functions (子程序和函数采用一个入口和一个出口)
- D. Limited use of pointers (限制使用指针)
- E. No unconditional jumps (没有无条件跳转)

6. Hardware Development (硬件开发)

Q36. Which distribution of failure modes is used for a ceramic capacitor according to A. Birolini, *Reliability Engineering — Theory and Practice*, 6th edition?

根据 A. Birolini 《可靠性工程：理论与实践》第 6 版，陶瓷电容使用哪种失效模式分布?

- A. Short 70%, Open 10%, Drift 20%, Functional — (短路 70%、断路 10%、漂移 20%、功能性 —)
- B. Short 15%, Open 80%, Drift 5%, Functional — (短路 15%、断路 80%、漂移 5%、功能性 —)
- C. Short 20%, Open 20%, Drift 50%, Functional 10% (短路 20%、断路 20%、漂移 50%、功能性 10%)

Q37. Which expressions are true?

下面哪些描述是正确的?

- A. FMEDA shows complex relations and interdependencies (FMEDA 显示复杂的关系和相互依赖性)
- B. Circuit diagrams and part lists are used as input for FMEDA (电路图和零部件清单用作 FMEDA 的输入)
- C. FMEDA is used for calculation of SPFM and LFM (FMEDA 用于 SPFM 和 LFM 的计算)
- D. For determination of failure rates, it is generally allowed to mix data sources; the best values of each source may be used (确定失效率时通常允许混合数据来源，并可使用各来源的最佳值)

E. Only one standard shall be used for determination of failure rates; mixing sources is allowed only in exceptional cases (应使用一个标准确定失效率; 只有在特殊情况下才允许混合不同来源)

Q38. Which methods can be used to calculate PMHF?

以下哪些方法可用于计算 PMHF 的值?

- A. DFA
- B. FMEDA
- C. Quantitative FTA (定量故障树分析)
- D. HARA

Q39. Which random-hardware-fault classes are mentioned in ISO 26262-5?

ISO 26262-5 中提到了哪些随机硬件故障类别?

- A. Dangerous fault (危险故障)
- B. Residual multiple-point fault (残余多点故障)
- C. Safe fault (安全故障)
- D. Detected multiple-point fault (被探测的多点故障)
- E. Perceived multiple-point fault (被感知的多点故障)
- F. Undetected multiple-point fault (未检测到的多点故障)
- G. Residual single-point fault (残余单点故障)
- H. Residual fault (残余故障)